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PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (*INFORMATION SHEET*)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-3:2015CORE ABILITIES	
TAHAP / LEVEL	3 (THREE)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA06 HEALTH, SAFETY AND ENVIRONMENTAL IMPLEMENT CONSCIOUSNESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	6.1 EXECUTE HEALTH CONSCIOUSNESS 6.2 EXECUTE SAFETY CONSCIOUSNESS 6.3 EXECUTE ENVIRONMENT COMPLIANCE	
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TAJUK/TITLE: SAFETY AND HEALTH SYSTEM MONITORING

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the Occupational Safety and Health (OSH) Act, Industrial Code of Practise (ICOP) and Guidelines. The person who is competent in Health, safety and environmental consciousness shall be able to execute health consciousness, execute safety consciousness and execute environment compliance. The person who is competent in this CA shall be able to execute his/her job functions in accordance to the requirements of health, safety and environment rules and regulations.

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PENERANGAN/INFORMATION:

1. Safety And Health Risk Control

The most important step in managing risks involves eliminating them so far as is reasonably practicable, or if that is not possible, minimizing the risks so far as is reasonably practicable. In deciding how to control risks you must consult your workers and their representatives who will be directly affected by this decision. Their experience will help you choose appropriate control measures and their involvement will increase the level of acceptance of any changes that may be needed to the way they do their job.

There are many ways to control risks. Some control measures are more effective than others. You must consider various control options and choose the control that most effectively eliminates the hazard or minimizes the risk in the circumstances. This may involve a single control measure or a combination of different controls that together provide the highest level of protection that is reasonably practicable.

Some problems can be fixed easily and should be done straight away, while others will need more effort and planning to resolve. Of those requiring more effort, you should priorities areas for action, focusing first on those hazards with the highest level of risk.

1.1 The Hierarchy Of Risk Control

The ways of controlling risks are ranked from the highest level of protection and reliability to the lowest as shown in Figure 1. This ranking is known as the hierarchy of risk control. The duty holders need to work through this hierarchy when managing risk.

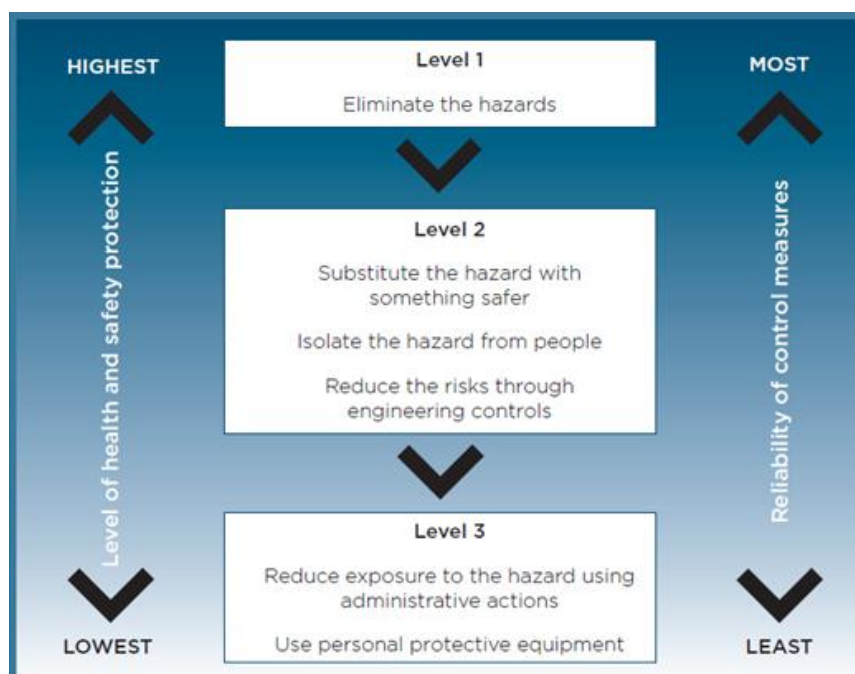


Figure 1: The hierarchy of risk control

You must always aim to eliminate a hazard, which is the most effective control. If this is not reasonably practicable, you must minimize the risk by working through the other alternatives in the hierarchy.

LEVEL 1 CONTROL MEASURES

The most effective control measure involves eliminating the hazard and associated risk. The best way to do this is by, firstly, not introducing the hazard into the workplace. For example, you can eliminate the risk of a fall from height by doing the work at ground level.

Eliminating hazards is often cheaper and more practical to achieve at the design or planning stage of a product, process or place used for work. In these early phases, there is greater scope to design out hazards or incorporate risk control measures that are compatible with the original design and functional requirements. For example, a noisy machine could be designed and built to produce as little noise as possible, which is more effective than providing workers with personal hearing protectors.

You can also eliminate risks by removing the hazard completely, for example, by removing trip hazards on the floor or disposing of unwanted chemicals. It may not be possible to eliminate a hazard if doing so means that you cannot make the end product

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or deliver the service. If you cannot eliminate the hazard, then eliminate as many of the risks associated with the hazard as possible.

LEVEL 2 CONTROL MEASURES

If it is not reasonably practicable to eliminate the hazards and associated risks, you should minimize the risks using one or more of the following approaches:

i) Substitute the hazard with something safer

For instance, replace solvent-based paints with water-based ones.

ii) Isolate the hazard from people

This involves physically separating the source of harm from people by distance or using barriers. For instance, install guard rails around exposed edges and holes in floors; use remote control systems to operate machinery; store chemicals in a fume cabinet.

iii) Use engineering controls

An engineering control is a control measure that is physical in nature, including a mechanical device or process. For instance, use mechanical devices such as trolleys or hoists to move heavy loads; place guards around moving parts of machinery; install residual current devices (electrical safety switches); set work rates on a production line to reduce fatigue.

LEVEL 3 CONTROL MEASURES

These control measures do not control the hazard at the source. They rely on human behaviour and supervision, and used on their own, tend to be least effective in minimizing risks. Two approaches to reduce risk in this way are:

i) Use administrative controls

Administrative controls are work methods or procedures that are designed to minimize exposure to a hazard. For instance, develop procedures on how to operate machinery safely, limit exposure time to a hazardous task, and use signs to warn people of a hazard.

ii) Use personal protective equipment (PPE)

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Examples of PPE include ear muffs, respirators, face masks, hard hats, gloves, aprons and protective eyewear. PPE limits exposure to the harmful effects of a hazard but only if workers wear and use the PPE correctly.

2. Method Of Supervision Work Instruction, Safe Operating Procedure And Emergency Response Plan

Supervision is a fundamental safety function applicable within all levels of an organisation. It complements the provision of information, instruction and training, and influences how well organisations achieve the safety and health objectives. Effective supervision sets and maintains high standards of performance and the physical aspects of the work environment, and is critical to achieving and maintaining the desired safety culture. Effective management and supervision can only be achieved through a multi-tiered, team approach where each level of the organisation plays a part in implementing the overall safe systems of work. Supervisors depend on direction and support from middle and senior management, as well as the involvement of the workers and contractors they direct.

Supervisors have in the area of occupational safety and health for all employees under their supervision establishing safety rules and safe working procedures. Train and instruct employees so they can perform their work safely. Know what personal protective equipment is needed for each task and how this equipment must be properly used, stored and maintained. When there are mandated safety training courses, ensure that your employees take them and that they are appropriately documented.

It's the supervisor's responsibility to enforce safe work practices and procedures; failure to do so is an invitation for accidents to occur. Workers must be encouraged to identify unsafe or unhealthful workplace conditions or hazards and absolutely not be disciplined for doing so.

3. Permit To Work System (PTW)

A PTW system is a formal written system used to control certain types of work which are identified as potentially hazardous. It is also a means of communication between site/installation management, plant supervisors and operators and those who carry out the work. Essential features of PTW system are:

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- i) clear identification of who may authorize particular jobs (and any limits to their authority) and who is responsible for specifying the necessary precautions
- ii) training and instruction in the issue and use of permits
- iii) monitoring and auditing to ensure that the system works as intended.

The terms “P.T.W”, “permit” or “work permit” refer to the certificate or form which is used as part of an overall system of work and which has been devised by a company to meet its specific needs. A P.T.W system aims to ensure that proper planning and consideration is given to the risks of a particular job. The permit is a written document which authorises certain people to carry out specific work, at a certain time and place, and which sets out the main precautions needed to complete the job safely.

The objectives and functions of such a system can be summarised:

- i) ensuring the proper authorization of designated work. This may be work of certain types, or work of any type within certain designated areas, other than normal operations
- ii) making clear to people carrying out the work the exact identity, nature and extent of the job and the hazards involved, and any limitations on the extent of the work and the time during which the job may be carried out
- iii) specifying the precautions to be taken including safe isolation from potential risks such as hazardous substances and energy sources
- iv) ensuring that the person in charge of a unit, plant or installation is aware of all the work being done there
- v) providing not only a system of continuous control but also a record showing that the nature of the work and the precautions needed have been checked by an appropriate person or people
- vi) providing for the suitable display of permits
- vii) providing a procedure for times when work has to be suspended, ie stopped for a period before it is complete
- viii) providing for the procedures or arrangements for work activities that may interact with or affect any of these activities
- ix) providing a formal hand-over procedure for use when a permit is issued for a period longer than one shift or when permit signatories change

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- x) providing a formal hand-back procedure to ensure that any part of the plant affected by the work is in a safe condition and ready for reinstatement.

3.1 Application of PTW System

Permits to Work should be considered whenever it is intended to carry out any work which may adversely affect the safety of personnel, the environment or the plant. They are normally considered to be more appropriate to non-routine activities which may require some form of Job Safety Analysis prior to work commencing. There will, however, be activities closely related to plant operations where P.T.W systems will be required. Maintenance work carried out by plant operators, for instance, should be subject to P.T.W procedures. It is also advisable to use a P.T.W system when two or more individuals or groups or people, perhaps from different trades or different contractors, need to co-ordinate their activities to ensure that their work is completed safely. This will apply equally when there is a transfer of work and responsibilities from one group to another. It is suggested that companies assess the risk of their activities and list specific operations and types of work which should be subject to P.T.W systems. It is not intended that P.T.W procedures be applied to all activities as experience has shown that their overall effectiveness may be weakened. It is very important for clear understanding by personnel moving from site to site, (especially contractors), that P.T.W systems are, as far as possible, harmonised between the different locations of the same Company. It is in any event essential that anybody starting work is familiar with the local instructions detailing when and how P.T.W systems are to be applied at a particular location.

3.2 Principal Responsibilities of PTW System

Several individuals/organisations will have specific duties which should be defined in the P.T.W. procedures. Principal responsibilities are identified below.

- i) Installation owners should ensure:
 - a) an appropriate P.T.W system is introduced
 - b) training programmes and competence standards are established and maintained

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- c) monitoring/auditing/reviewing of the P.T.W. system is established and maintained
- ii) The Installation Manager should ensure that:
 - a) all personnel who operate and use the P.T.W. system are competent to do so
 - b) the planning, issue and return of permits is properly co-ordinated
 - c) a secure method of electrical and process/mechanical isolations is implemented
 - d) adequate time is allowed during shift changes to ensure effective transfer of information on outstanding permits
 - e) the system is regularly monitored to ensure that the P.T.W system is implemented effectively.
- iii) Contractor's management should ensure that:
 - a) they are informed of and understand the broad principles of the P.T.W system for the locations where their employees are to work
 - b) their employees have been given the appropriate training and understand the operation of the P.T.W system and their specific responsibilities within it
 - c) they monitor the training of their employees.
- iv) The person who issues a permit should ensure that:
 - a) the nature of the work is fully understood
 - b) all the hazards associated with the job are identified
 - c) all the necessary precautions are implemented, including isolations, before work begins
 - d) all people who may be affected by the work are informed before the work begins, when the work is suspended and when the work is complete
 - e) permits for tasks that may interact are cross-referenced
 - f) that effective arrangements are made for the work site to be examined before work begins, on completion of work and as appropriate when work is suspended

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- g) sufficient time is spent on shift handover to discuss all ongoing or suspended permits with the oncoming permit issuer.
- v) Supervisors (or person in charge of the work) should ensure that:
 - a) they have received training in the P.T.W system as applied in that particular location
 - b) the people working for them have received adequate instruction in the system
 - c) they discuss the job fully with the person issuing the permit
 - d) the permit is posted at the work site
 - e) the work party is briefed on the details of the permit including any potential hazards, and on all the precautions taken or to be taken
 - f) the precautions are maintained throughout the work activity
 - g) the worker understands that if circumstances change work must be stopped and advice sought
 - h) the work group stays within the limitations set on the permit (physical boundaries, type of work and validity time)
 - i) on completion or suspension of the work the site is left in a safe condition and the permit issuer is informed.
- vi) Individuals working within the P.T.W system should ensure that:
 - a) they have received instruction and have a good understanding of the P.T.W. system at the installation where they work
 - b) they do not start any work requiring a permit, until it has been properly authorised and issued
 - c) they receive a briefing from the supervisor on the particular task and they understand the hazards and the precautions taken or to be taken
 - d) they follow the instructions specified in the permit
 - e) when they stop work, the site and any equipment they are using is left in a safe condition
 - f) if in any doubt or if circumstances change, they must stop work and consult with their supervisor.

3.3 Considerations When Using A Safe Work Permit

A safe work permit should only be issued by a competent person who is completely familiar with the work or situation covered by the permit and who has control over changes in that work area e.g. lead operator or supervisor. The permit issuer must be sure that the identified on the permit is as described. Where possible, the permit issuer should review the work or operation with the worker before work begins. If the permit issuer has not reviewed the site, this should be noted on the permit and the work situation should be discussed with the worker. Written instructions alone are often insufficient in the effective use of a permit system.

Practical training exercises for the people who issue and receive permits should be considered. The person receiving the permit must completely understand the work situation, the potential hazards, and the precautions required before accepting the permit. Any special precautions not normally associated with the particular work should be identified to the receiver of the permit, who must fully understand the reasons for these precautions e.g. work to be done in an area where there is a possible exposure to H₂S gas.

The permit issuer must be sure that the worker understands the hazards. If not, the permit issuer needs to review the Material Safety Data Sheet or other information with the worker to ensure that they understand the dangers of the product and the precautions to be taken. No one should sign a safe work permit unless completely satisfied that the work can be done safely. All safe work permits must be signed by both the permit receiver and the permit issuer before work is started and after it is completed.

3.4 Pitfalls Of Work Permits

Workers and supervisors do not always see the need for a safe work permit system. They have not been trained to recognize the added safeguards that such a program provides.

Factors leading to ineffective work permit systems are:

- i) The type or format of the permit does not cover all the potential hazards.
- ii) The issuing procedure is inadequate.

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- iii) The person signing the permit has not inspected the operation to see if the isolation, lock-out, or testing has been done.
- iv) Workers are not following or don't understand the requirements of the permit, especially the expiry time.
- v) The employer is not enforcing or auditing the work permit system.
- vi) Permits are prepared too far in advance, or after the work have begun.
- vii) A responsible person is not inspecting the operation after the permit has been issued.
- viii) The system is too complex.

All workers using permits must completely understand the reasons for, and the requirements of, the permit before work begins. A safe work permit is an effective tool to help identify and control hazards, prevent injuries, and avoid costly mistakes.

4. Workplace Inspection Checklist

Effective safety and health inspections are one of the most important incident/accident prevention tools in a company's safety and health program. Using properly trained inspectors in a planned inspection program will reduce incidents and property damage. An effective safety inspection program will improve worker communication, company morale and, over time, save the employer money.

Too often, safety inspections are aimed primarily at finding and recording unsafe conditions. This narrow focus tends to ignore other causes of incidents, such as unsafe actions and personal factors. In addition, workers and supervisors are generally well aware of the inspection teams arrival a day or two before the inspection. This warning system sometimes creates a preparatory atmosphere before the inspectors arrive. That means the inspectors often observe the workplace and those within it only on a superficial basis. The result is that safety inspectors rarely see the actual situations that are causing the incidents, injuries and property damage. In order for your inspection team to be effective, they must inspect the workplace in its day-to-day status. They must see the activities and the conditions in which incidents, injuries and property damage occur.

An inspection program takes planning, preparation and training. For inspectors to do a good job, they must receive training in what to look for and what to look at. There must be a guidance system in place to maintain consistency of inspections. A provision for

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recording and reporting any deficiencies to management, the safety committee and supervisors. The last part of the system should provide a means of follow up and the monitoring of any deficiencies identified.

4.1 Preparing For An Inspection

Safety and health inspection programs require planning. It is important that employers have adequate policies and procedures in establishing their safety inspection programs. Responsibility and accountability must be assigned, identifying who inspectors are and when inspections will be done. The primary focus of this program should be accident prevention, through the maintenance of safe working conditions and the removal of any potential hazards that arise in the workplace.

Good inspection programs will also identify the items to be inspected and then set standards to be maintained in the workplace by supervision and workers. The program should include a system that will record inspections done and ensure that any problems identified are corrected by a responsible person who has the ability to carry out the necessary changes. A follow-up system, through the safety committee and management, should be employed to ensure that all items are followed up and corrected. Once the inspection program is in place, the safety program should provide a means of monitoring it to see if trends arise that may be contributing to the company's accident or injury problems.

4.2 Meeting The Legal Requirement

The Safety and Health Committee Regulations 1996 applies to all employers and safety and health committee. Regulation 12 of this regulation require a safety and health committee shall inspect the place of work at least once in every 3 months to ascertain if there is anything prejudicial to the safety & health of persons employed therein. Discuss the observations and keep proper records. Make recommendations to the employer on the remedial measures.

4.3 Developing an Inspection Program

The regulations tell us that we are to inspect for 'hazardous conditions, unsafe actions or work methods'. These are only the general requirements and will require further detail from employers identifying the safe conditions, actions and work methods for their workplace.

Employers must develop their own standards and procedures of work to meet the requirements of the applicable regulatory, industry, and manufacturers' standards for their workplace. They are to develop procedures of work methods for hazardous jobs that their workers may encounter such as lockout, confined space work, emergency procedures and so on. Employers must maintain safe working conditions and practices.

The extent to which a person can carry out an effective inspection depends upon their ability to identify hazards. Inspectors should be provided with a reference to follow during their inspections.

4.4 Using Checklists

Many companies use a checklist to ensure consistent and comprehensive inspections each time they are done. Appropriate checklists are developed for each job site and provide a guide to the various standards expected to be in place. A Checklist should inform inspectors what to look at and what to look for.

i) What should we be looking at?

- Consider the following in the development of your checklists and guidelines:
- plant or job site layout (areas where work activities take place)
- building-structure
- basic floor plan layouts with equipment and machinery
- maintenance periods-shift work
- start up and shut down times
- hazardous substances used in the workplace
- storage areas exits

ii) What are we looking for?

Operating standards or requirements within these areas:

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- legal requirements (OH&S Regulation, fire regulations, boiler pressure vessel, elevating devices etc.)
- company rules/regulations
- manufacturers' specifications and instructions (forklifts, maintenance and operating procedures)
- personal protective equipment required (headgear, footwear, gloves, respirators, locks)
- engineering controls(ventilation, guards etc.)
- emergency procedures (fire, evacuation etc.)
- first aid services and supplies.

Look at known problem areas and review records on the following:

- accident investigations
- first aid record books
- worker complaints and reports on hazards in the workplace
- recommendations made by safety and health committees
- previous inspections (including fire marshal, etc.)
- maintenance reports
- what controls are used to ensure safe work practices and maintain conditions?

Workplace floor plans should be referenced during the planning stages of any checklist. They should include information that will indicate locations of previous incidents, hazardous work areas, health hazard areas, and any relative information that would assist in the development of a checklist.

4.5 Monitoring

Information obtained from your inspection reports should be reviewed and become part of your OH&S program records and statistics. Joint health and safety committees

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and Worker health and safety representatives should review each inspection report to identify any trends that may be developing in the workplace. A proper analysis over time may reveal:

- a need for training in certain areas
- why incidents are occurring in certain areas
- the need to establish priorities for corrective action
- a need to develop or improve safe work practices
- problem areas that may require more hazard analysis.

5. Toolbox Meeting

Toolbox meetings are a conduit for information and provide workers with the opportunity to have their say about hazards/controls, incidents/accidents, work processes and company procedures. Toolbox meetings should be run on a regular basis; the frequency of meetings will depend on the size, nature and location of the site. Some hazardous activities could require daily meetings, while often a weekly/fortnightly meeting will suffice.

5.1 Why do we run Safety Meetings?

- i) Inform workers of changes to company procedures
- ii) Identify new hazards and review existing hazards
- iii) Develop/review hazard controls
- iv) Discuss/review accident and incident data
- v) Employee participation
- vi) Communication
- vii) Discuss programme
- viii) Develop/review work processes
- ix) Short training sessions.

5.2 Structure of Meetings

Safety meetings for workers should be short and to the point. Meeting on a regular basis and running for 10-15 minutes is an ideal way to involve workers and make the most effective impact. Monday morning is an ideal time to run the meeting as workers can be briefed on the weeks programme and review the previous week's activities. Depending on the nature of the business it is advisable to conduct an inspection of the work area prior to a meeting to help identify any problems and to assist with topical items for discussion. Details of meetings should be recorded and kept on file. It is important to record meeting dates, attendees and discussion items, even follow up items from hazards, accidents and incidents. Managers and supervisors will normally run the safety meeting but workers can also assist in the process. It is a good idea to use trainees or young workers to conduct inspections and run meetings as they are the future managers, they are not biased by entrenched behaviours and it gives them a good grounding in the hazard management process.

5.3 Topics for Discussion

Discussion points at safety meetings should be topical and relevant to current or upcoming activities in the workplace. Topical items for discussion can be identified by asking workers for input, changes in the plant or work process or work environment, or in response to accidents/incidents in the workplace.

Listed below are some suggested topics for discussion:

- i) What items would workers like to discuss?
- ii) Introduction of new plant or processes
- iii) Development of task analysis or methodologies
- iv) Changes in season e.g. sun smart, dehydration
- v) Use of plant
- vi) Handling of materials
- vii) Identifying training requirements.

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SOALAN/QUESTION :

1. Explain the risk control hierarchy

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2. What are essential features of PTW system?

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3. List the causes of work permits to be ineffective

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RUJUKAN / REFERENCES:

1. Factories and Machinery Act 1967 and Regulations, Pindaan Ogos 2008, ISBN 967701156-1
2. Occupational Safety and Health Act 1994 and Regulation, (2009) Malaysia, Penerbitan Akta (M), ISBN-967701188-X.
3. Garis Panduan bagi Pengenalpastian Hazard, Penaksiran Risiko dan Kawalan Risiko (HIRARC), ISBN 978-983-201